MIL-F-52393A(EL)
4 June 1971
Superseding
MIL-F-52393(MO)
14 October 1964

MILITARY SPECIFICATION
FILTER KIT, INFRARED
FOR GENERAL PURPOSE VEHICLES

1. SCOPE

1.1 Scope. - This specification covers three types of infrared filters for mounting on general purpose vehicles.

1.2 Classification. - Filters shall be classified as follows:

Type I	- Glass
Type II	- Molded plastic
Type III	- Composite

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	- Pallet, Materials Handling, Wood, General Construction Requirements
QQ-S-781	- Steel, Strapping, Flat
PPP-B-585	- Box, Wood, Wire-bound
PPP-B-601	- Boxes, Wood, Cleated - Plywood
PPP-B-621	- Box, Wood, Nailed and Lock-Corner
PPP-B-636	- Box, Fiberboard
PPP-B-640	- Box, Fiberboard, Corrugated, Triple Wall
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	- Strapping, Nonmetallic, and Connectors
PPP-T-76	- Tape, Pressure Sensitive Adhesive Paper, Water Resistant, for Carton Sealing

FSC 5855

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PPP-T-97	- Tape, Pressure Sensitive Adhesive Filament Reinforced

Military

MIL-P-116	- Preservation, Methods of
MIL-C-4150	- Case, Carrying, Water-Vaporproof

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U. S. Military Property
MIL-STD-147	- Palletized and Containerized Unit Loads, 40 Inches x 48 Inches, Pallets Skids, Runners, or Pallet Type Base
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts

DRAWINGS

ECOM

SC-DL-340630	- Filter Kit, Infrared for General Purpose Vehicle

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The infrared filter kit shall consist of: two filter assemblies for flush headlamps; two neoprene adapters for pedestal headlamps; two clamp assemblies for pedestal headlamps; two filter assemblies for taillamps; eight cover assemblies, blackout, for clearance lights; six 10 x 24 and six ¼ x 20 machine screws; and a carrying case. The infrared filter kit shall hereinafter be known as "kit".

3.2 Construction. - The kit shall be in accordance with SC-DL-340630 and as specified herein.

3.2.1 Weight. - The kit shall weigh not more than 17 pounds.

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3.3 Preproduction. - The contractor shall furnish kits (see 6.2) within the time frame specified in the contract to prove, prior to starting production, that his production methods and choice of design detail will produce kits which comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the contractor of the obligation to provide kits conforming to this specification.

3.4 Initial production. - When specified (see 6.2), the contractor shall furnish to the Government kits for inspection as specified in 4.4.

3.5 Material and components. - Material and components shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.6 Performance and environmental.

3.6.1 Thermal shock. - The kits shall operate as specified herein and shall not be damaged (see 6.3.1) when tested as specified in 4.6.1.

3.6.2 Effective visual transmission (EVT). - The EVT (see 6.3.2) of the filters shall be 5×10^{-9} maximum for the headlamp filters and 1×10^{-8} maximum for the taillamp filters.

3.6.3 Effective holo transmission (EHT). - The EHT (see 6.3.3) of the filters shall be 13 percent minimum for the headlight filters and 15 percent minimum for the taillamp filters.

3.7 Carrying case. - The carrying case shall conform to MIL-C-4150 and shall be as shown in the drawings.

3.8 Identification markings. - Identification markings shall be in accordance with MIL-STD-130 and as shown on the drawings.

3.9 Workmanship. - Filters shall show no evidence of chips, cracks, striations, or any imperfection which might impair their optical qualities or mechanical strength. All other parts and components shall fit tight. Edges and mating surfaces shall be smoothly finished and shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (see 4.3).

b. Initial production inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Inspection comparison (see 4.7).

e. Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. - Each kit shall be examined for the following defects. One or more defects shall be cause for rejection.

101. Dimensions or weight not as specified.

102. Absence of any part or component.

103. Fit of parts or components not as specified.

104. Materials or components not as specified.

105. Carrying case not in accordance with MIL-C-4150.

106. Treatment and painting not as specified.

107. Incorrect or missing identification markings.

108. Workmanship not as specified.

4.3.2 Tests. - Upon successful completion of the examination specified in 4.3.1, the preproduction model shall be tested in accordance with 4.6.1 through 4.6.3.

4.4 Initial production inspection. - When specified (see 3.4), the initial production units will be selected at random by the Government from the first kit production lot, shall be examined as specified in 4.3.1 and tested as specified in 4.3.2. to determine conformance to the requirements of this specification. The inspection will be performed by the contractor under supervision of a representative of the contracting officer. Acceptance of initial production kits shall not exclude the remaining kits from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. - Failure of an initial production kit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4. shall be cause for refusal by the Government to accept production kits until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies.

4.5 Quality conformance inspection.

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4.5.1 Sampling. - Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-3. AQL for examination and tests combined shall be in 6.5 percent defective.

4.5.2 Examination. - Samples selected in accordance with 4.5.1 shall be examined in accordance with 4.3.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Group A. - Each filter shall be tested in accordance with 4.6.2 and 4.6.3. Failure of either test shall be cause for rejection.

4.5.3.2 Group B. - Sample filters selected in accordance with 4.5.1 and which have passed the tests specified in 4.5.3.1 shall be tested in accordance with 4.6.1, 4.6.2 and 4.6.3 in that order.

4.6 Test procedure. - Unless otherwise specified all tests shall be performed at room temperature (see 6.3.4). Tests shall be performed on the filters only.

4.6.1 Thermal shock. - The filters shall be tested as specified in MIL-STD-202, Method 107B. Test condition B shall apply to type I and II filters and test condition A shall apply to type III filters. Any evidence of damage (see 6.3.1) shall constitute failure of this test.

4.6.2 Effective visual transmission (EVT). - A high density filter photometer constructed as described in 6.4 and shown in figure 1 is used to measure EVT. The photometer should be calibrated by comparison against a record standard lamp in order to correct for losses in the S_2 standard lamp optical path caused by the opal glass and matched filters. The ribbon filament lamp is adjusted for a color temperature of 2870° K, and its energy output determined. The standard lamp S_2 is then varied to obtain a brightness match and color match is obtained by insertion of one of the color matched filters. EVT is then the ratio of the energy output from the standard lamp S_2, opal glass, color matched filter combination to the energy output from the ribbon filament lamp. Failure to comply with the values specified in 3.6.2 shall constitute failure of this test.

4.6.3 Effective holo transmission (EHT). - EHT test equipment as defined in 6.5 and shown in figure 2 shall be used. A 100 percent reading is taken by reading the output current of the photomultiplier with white light from S falling on the cathode of the image tube. The filter under test is then interposed at aperture A and the output is again measured. The ratio of output with the filter to output without the filter is the EHT. Failure to comply with values specified in 3.6.3 shall constitute failure of this test.

4.7 Inspection comparison. - The Government may select kits at any time during the contract production period and subject the kits to the examinations and tests specified for the initial production kit to reveal any manufacturing deficiencies. The tests will be performed by the Government at a site selected by the Government. Kits selected for inspection comparison will not include the previously selected and inspected preproduction or initial production kits.

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4.7.1 Inspection failure. - Failure of any kit when examined and tested as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of kits until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection.

4.8.1.1 Unit of product. - For the purpose of inspection, a kit prepared for shipment, shall be considered a unit of product.

4.8.1.2 Sampling. - Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. - Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109. Packaging and packing not in accordance with 5.1 and 5.2.

110. Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. - Infrared Filter Kit, MK-1159/PAS-5, shall be cleaned in accordance with C-1 of MIL-P-116.

5.1.1.2 Drying. - Infrared Filter Kit, MK-1159/PAS-5, shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Infrared Filter Kit, MK-1159/PAS-5. - Each Infrared Filter Kit, MK-1159/PAS-5, in its transit case shall be individually packaged method III as follows: Cushion the transit case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, variety SW, grade W5c, class weather-resistant designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, variety SW, grade W5c. Place the technical literature packaged as specified in

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5.1.1.4.1, on top of the contents, directly under the lid of the container. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. - Infrared Filter Kit, MK-1159/PAS-5 in its transit case shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents.

5.2.1 Level A.

5.2.1.1 Palletized load. - A quantity of Infrared Filter Kits, MK-1159/PAS-5, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be of softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which would adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto, except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. - A quantity of Infrared Filter Kits, MK-1159/PAS-5, packaged as specified in 5.1 shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Infrared Filter Kits, MK-1159/PAS-5, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specifi-

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cation. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. - A quantity of Infrared Filter Kits, MK-1159/PAS-5, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Infrared Filter Kits, MK-1159/PAS-5, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The filter kit, infrared, is intended to provide infrared driving illumination from existing head and taillamps on general purpose vehicles for night driving when used with the binocular, electronic, AN/PAS-5.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Number of preproduction and initial production filter kits required.

c. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Definitions.

6.3.1 Damage. - Damage is defined as chipping, breaking, pitting, cracking, or peeling of the filters.

6.3.2 Effective visual transmission (EVT). - EVT is that fraction of the total available visible light which is transmitted by the filter.

$$EVT = \frac{\int_0^\infty J_\lambda K_\lambda T_\lambda d\lambda}{\int_0^\infty J_\lambda K_\lambda d\lambda}$$

where J_λ is the spectral distribution of a tungsten light source operating at a color temperature of 2870° K, K_λ is the spectral response or luminosity function of the dark-adapted eye, T_λ is the spectral transmission of the filter in question, and λ is the wave-length of the radiation.

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6.3.3 Effective holo transmission (EHT). - EHT is that portion of the total available infrared energy which is transmitted by the filter.

$$EHT = \frac{\int_c^\infty J_\lambda R_\lambda T_\lambda d\lambda}{\int_o^\infty J_\lambda R_\lambda d\lambda}$$

where R_λ is the spectral response of the infrared sensitive cathode of the detection device and the other quantities are those defined in 6.3.2.

6.3.4 Room temperature. - Room temperature is defined as 23° C ± 3° C.

6.4 EVT test equipment. - An optical diagram of a high-density filter photometer used for measuring EVT is shown in figure 1. S_1 is a high current ribbon filament lamp which is focused by lenses L_1 and L_2 onto the plane of the filter under test at aperture A_1. S_2 is a small 6-candlepower (cp) standard lamp. A standard lamp is calibrated by the National Bureau of Standards and includes a calibration curve of candlepower vs. current. The light from this source is diffused by an opal glass and passes through one of four selectable color-matching filters to the Lummer-Brodhun cube. The Lummer-Brodhun cube consists essentially of two right angle prisms, theouter portions of the hypotenuse of one being ground away before they are cemented together. When illuminated from two sources, as shown in figure 1, the result is a field composed of three vertical strips presented to the eyepiece L_3 and aperture A_2. When the current to the standard lamp is varied such that the intensity of the standard lamp is made to equal the unknown source intensity, the unknown may be calculated from the standard lamp calibration curves of candlepower vs. current and a linear proportion involving the square law of the distances.

NOTE: Of the standard sources, the tungsten ribbon-filament lamp is generally used for the near-infrared source applications. It should be noted, however, that for a color operating temperature of 2800° K, approximately 50 amperes is required. The life-time of the tungsten lamp at this elevated temperature is relatively short and external cooling is required.

6.5 EHT test equipment. - A diagram of the optical system of an infrared filter photometer used in measuring EHT appears in figure 2. S is a tungsten source calibrated for 2870° K color temperature. L_1 is a lens for focusing the light through aperture A and onto the cathode R of an image tube having Electronic Industries Association (EIA) Standard S-1 sensitivity. L_2 is a lens suitable for focusing the output of the image converter tube onto a photomultiplier tube P with EIA S-4 sensitivity.

CUSTODIAN:
Army-EL

PREPARING ACTIVITY:
Army-EL

Project No: 5855-A075

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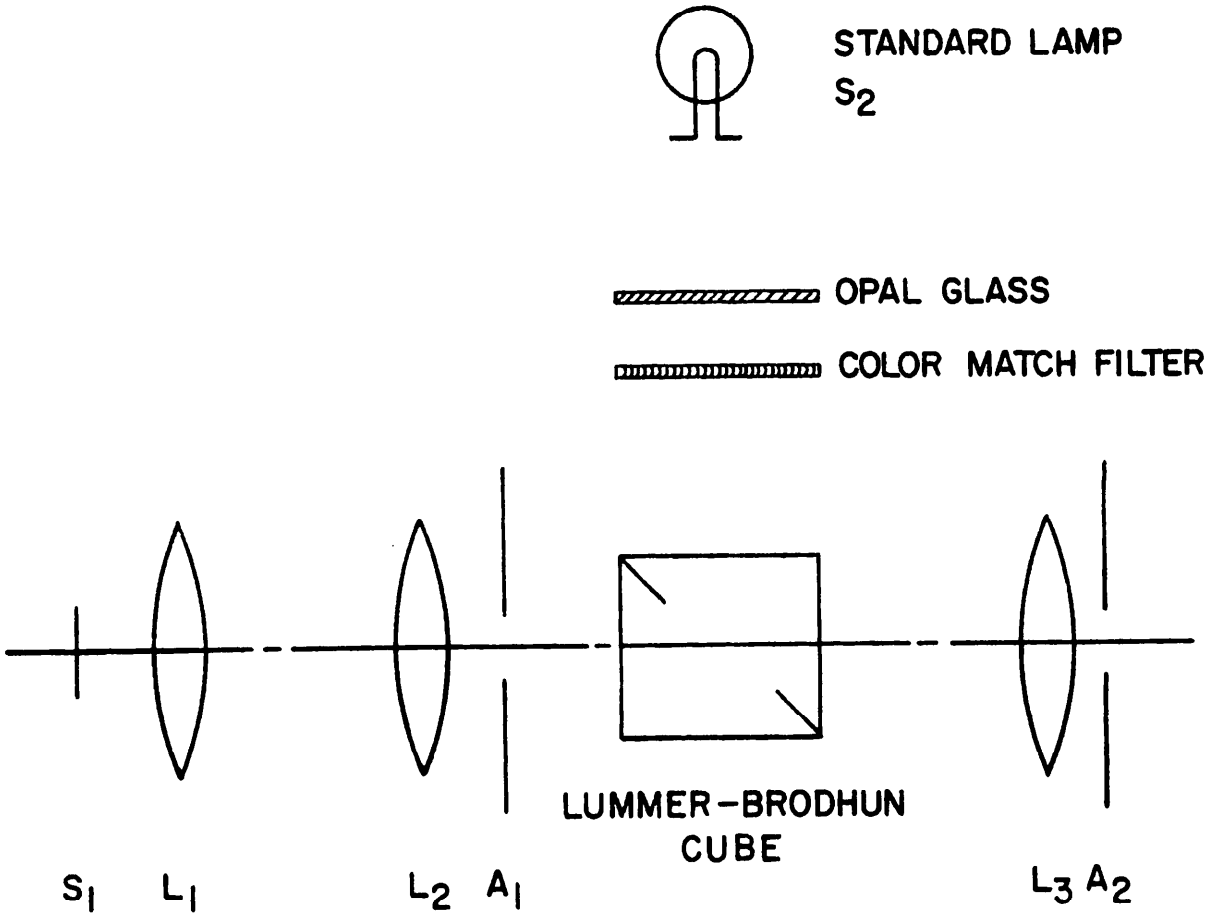


FIGURE 1. OPTICAL DIAGRAM OF EVT PHOTOMETER.

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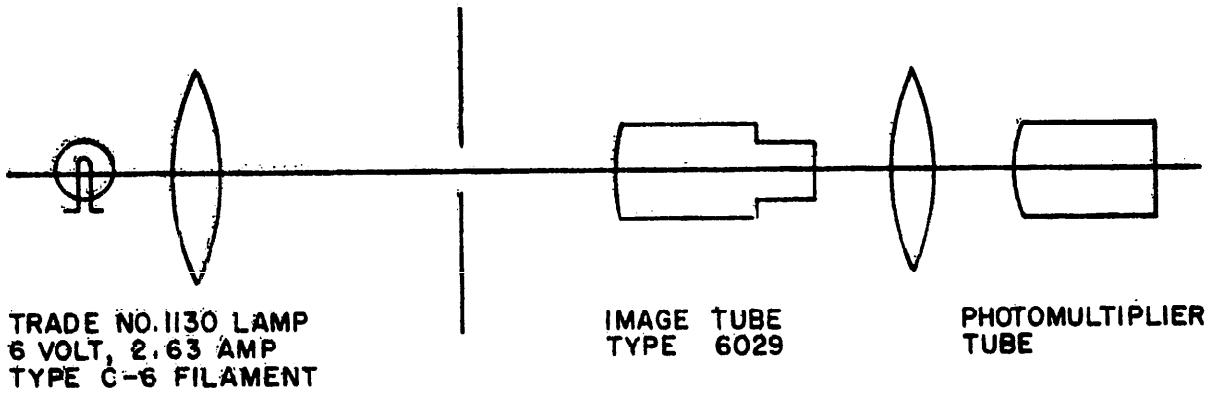


FIGURE 2. OPTICAL DIAGRAM OF EHT PHOTOMETER.

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-F-52393A(EL) FILTER KIT, INFRARED FOR GENERAL PURPOSE VEHICLES

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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